

Lorentzian Boundary Kernels as a Compactification Test for Primordial Cosmology

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We formulate a compactification test based on the constrained Lorentzian boundary kernel of the Feynman–DeWitt wave functional. At quadratic order its imaginary part is the inverse covariance of the physical boundary perturbations, so scalar, tensor, and isocurvature spectra and stability conditions follow from one matrix-valued object. Heavy moduli and Kaluza–Klein modes enter through a Schur complement, exposing their decoupling or observable imprint before a full Boltzmann analysis. We recover the standard single-field result and apply the construction to $M_4 \times S^2 \times \mathbb{CP}^2$, including the (3,3) and (1,5) flux branches, and to $M_4 \times S^3 \times S^3$ as a control. The method is an acceptability test, not a unique reconstruction of the internal manifold.

I. INTRODUCTION

The cosmic microwave background (CMB) constrains the statistics of primordial curvature, tensor, and isocurvature perturbations rather than the microscopic theory that produced them. For a conventional single-field model, the route from an inflationary background to these statistics is well established. For models with additional scalar fields, compactification moduli, Kaluza–Klein (KK) modes, nontrivial field-space geometry, or modified gravitational sectors, the same route becomes substantially less transparent. The observable perturbation can mix with hidden degrees of freedom before horizon exit, and the quantities ultimately supplied to a Boltzmann code are then entries of a coupled primordial covariance matrix rather than a single power law.

The practical problem addressed here is therefore not to replace existing CMB analysis software. Astrophysicists already have mature tools for evolving primordial spectra through recombination and for comparing the resulting angular spectra with microwave-background data. What is often missing is a compact theory-side interface between a coupled early-universe model and those established pipelines. A useful method should accept the background and quadratic perturbation matrices of a model, evolve them independently for each comoving wavenumber k , return scalar, tensor, and isocurvature covariances, and make clear when additional sectors have decoupled or have left an observable imprint.

The relevant cosmological sequence must also be kept distinct. A quantum or compactification boundary need not coincide with the onset of inflation, and the onset of inflation need not coincide with the horizon exit of the modes observed in the CMB. Schematically,

quantum boundary $\longrightarrow$ pre-inflation
 $\longrightarrow$ inflation $\longrightarrow k = aH \longrightarrow$ CMB.

The scalar amplitude is fixed by the background at horizon

exit,

$$\Delta_\zeta^2(k_*) \simeq \frac{H_*^2}{8\pi^2 \epsilon_* \bar{M}_P^2}, \quad (1)$$

not by the Hubble scale at the earliest admissible boundary. An intermediate expanding phase can separate the two scales. For example, if that phase is radiation-like, $H \propto a^{-2}$ and

$$\Delta N_{\text{pre}} = \frac{1}{2} \ln \left(\frac{H_Q}{H_{\text{inf}}} \right). \quad (2)$$

Taking $H_Q \simeq 0.84 M_P$ and $H_{\text{inf}} \simeq 3 \times 10^{-6} M_P$ gives about 6.3 pre-inflationary e-folds. This estimate is illustrative rather than a prediction; the equation of state and the transition into inflation must be supplied by the model being tested.

We formulate the missing interface in terms of the quadratic Lorentzian boundary kernel of the Feynman–DeWitt wave functional. After the lapse, shift, gauge redundancy, and nondynamical matter variables have been eliminated, the kernel is a matrix on the physical perturbation space. Its Hermitian imaginary part is the inverse Gaussian covariance of the boundary state. Thus one matrix contains the information required for the primordial scalar, tensor, and isocurvature spectra. Hidden fields are removed by a Schur complement, making their decoupling, instability, or residual scale dependence explicit before any transfer-function calculation is performed.

At quadratic order this is standard cosmological perturbation theory organized around a common computational object, not an alternative quantum theory. For a fixed background, contour, and Gaussian state it is equivalent to canonical mode evolution, the Schrödinger wave functional, and the in-in formalism [1–3]. Its advantage is organizational: the same matrix evolution handles adiabatic, entropy, tensor, modulus, and retained KK sectors, while positivity of the imaginary kernel provides a direct diagnostic of whether the chosen saddle and state define an acceptable covariance.

The output of the calculation is a tabulated primordial covariance on a k grid. In the adiabatic limit this reduces to $\mathcal{P}_{\mathcal{R}}(k)$ and $\mathcal{P}_T(k)$. With additional light sectors

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it also includes isocurvature spectra and cross correlations. These quantities can be passed to an existing CLASS-, CAMB-, or equivalent Boltzmann workflow; the subsequent computation of C_ℓ^{TT} , C_ℓ^{TE} , C_ℓ^{EE} , lensing, and likelihoods is unchanged. The boundary kernel itself is not an observable measured by Planck or any other CMB experiment. It is the theory-side object from which the primordial inputs to those analyses are derived.

We illustrate the method with two ten-dimensional products. The principal worked example is

$$\mathcal{M}_{10} = M_4 \times S^2 \times \mathbb{CP}^2, \quad (3)$$

used in the companion UECKK construction, its zero-mode family analysis, and its spin-torsion stability study [4–6]. Its two radii and quantized fluxes provide a nontrivial coupled modulus sector. The branches $(m, n) = (3, 3)$ and $(1, 5)$ both satisfy

$$N_{\text{fam}} = m \frac{n^2 - 1}{8} = 3, \quad (4)$$

but distribute flux differently and therefore have different canonical masses and potential CMB responses. We also use $M_4 \times S^3 \times S^3$ as a geometrically distinct control. The numerical examples are intended to show how a cosmologist connects model data to the kernel engine and then to existing CMB software. They are screening calculations, not a claim that present microwave-background data uniquely identify either internal manifold.

The paper is organized as follows. Section II defines the constrained boundary kernel and its covariance. Section III connects it to primordial observables. Section IV specifies the compactification data required by the calculation. The remaining sections develop the Schur-complement reduction, the worked compactification examples, and the per- k numerical implementation, and then describe the handoff to a full CMB likelihood analysis.

II. THE CONSTRAINED LORENTZIAN BOUNDARY KERNEL

A. Wave functional and saddle width

Let $q_A(\mathbf{x})$ denote a complete set of physical boundary variables on a spacelike hypersurface Σ . Before reduction, these variables are drawn from the induced spatial metric, matter fields, and internal metric components. The Feynman–DeWitt wave functional is

$$\Psi[q] = \int_{Q|\Sigma=q} \mathcal{D}Q \exp\left(\frac{i}{\hbar} S[Q]\right), \quad (5)$$

where Q collectively denotes bulk geometries and matter histories. The boundary data q are not the bulk integration variables. They label the state obtained after the bulk histories have been summed.

Suppose the integral is dominated by a saddle $\bar{Q}$ and write $Q = \bar{Q} + \eta$. After gauge fixing and elimination of constraints, the action has the quadratic expansion

$$S[\bar{Q} + \eta] = S[\bar{Q}] + \frac{1}{2} \eta \cdot \mathcal{A} \cdot \eta + \dots \quad (6)$$

The contour that defines the state selects a generally complex solution of the linearized equations. Evaluating the quadratic action on that solution gives a boundary term,

$$S_{\text{os}}^{(2)}[q] = \frac{1}{2} \int \frac{d^3k}{(2\pi)^3} q_A(-\mathbf{k}) K_{AB}(k) q_B(\mathbf{k}). \quad (7)$$

The matrix K_{AB} is the Dirichlet-to-Neumann map: it converts the boundary amplitude into the canonical momentum of the regular bulk solution,

$$K_{AB}(k) = \left. \frac{\delta \Pi_A(\mathbf{k})}{\delta q_B(\mathbf{k})} \right|_{\Sigma}. \quad (8)$$

Write $K = K_R + iK_I$, with K_I Hermitian. Then

$$|\Psi[q]|^2 \propto \exp\left[-\frac{1}{\hbar} \int \frac{d^3k}{(2\pi)^3} q_A(-\mathbf{k}) (K_I)_{AB} q_B(\mathbf{k})\right]. \quad (9)$$

A normalizable Gaussian requires K_I to be positive definite on the physical subspace. The covariance is

$$\langle q_A(\mathbf{k}) q_B(\mathbf{k}') \rangle = (2\pi)^3 \delta^3(\mathbf{k} + \mathbf{k}') \frac{\hbar}{2} (K_I^{-1})_{AB}. \quad (10)$$

Equation (10) is the central diagnostic of this paper.

The saddle is the mean history and the quadratic neighborhood is its quantum width. The contour and Hessian determine the associated uncertainty ellipse for the full constrained field configuration, not independent random values in spatial cells.

B. Constraints and physical variables

For gravity the ADM action has the form

$$S = \int dt d^3x \left[\pi^{ij} \dot{h}_{ij} + p_I \dot{\phi}^I - N\mathcal{H} - N^i \mathcal{H}_i \right]. \quad (11)$$

The lapse and shift are Lagrange multipliers. Their functional integration enforces the Hamiltonian and momentum constraints. At the linearized level these constraints remove scalar and vector gauge directions and relate matter and metric perturbations. A practical calculation may solve the constraints before the path integral or integrate over nondynamical variables directly; both procedures give the same reduced kernel when the measure is treated consistently.

For an isotropic four-dimensional background with scalar fields φ^a , a convenient physical basis is

$$q_A = (\zeta, s^\alpha, h_+, h_\times, \chi^r), \quad (12)$$

where ζ is the adiabatic curvature perturbation, s^α are isocurvature directions, h_λ are the two tensor polarizations, and χ^r are retained internal or KK modes. The kernel in this basis is gauge invariant. A negative eigenvalue of K_I is not a coordinate instability; it signals that the selected contour does not define a normalizable state in that direction or that the saddle is unstable.

C. Moving the boundary

The division between an “initial state” and “later mode evolution” depends on where an intermediate hypersurface is placed. Let $G_{21}(q_2, q_1)$ be the quadratic propagator between two boundaries. Composition gives

$$\Psi_2(q_2) = \int \mathcal{D}q_1 G_{21}(q_2, q_1) \Psi_1(q_1). \quad (13)$$

For Gaussian kernels this integration is exact. If

$$S_{21}^{(2)} = \frac{1}{2} q_2 A q_2 + q_2 B q_1 + \frac{1}{2} q_1 C q_1 \quad (14)$$

and the earlier state has kernel K_1 , then, up to convention-dependent factors of i ,

$$K_2 = A - B(C + K_1)^{-1} B^T. \quad (15)$$

Moving the intermediate boundary changes K_1 and the propagator separately but not the final state. One may therefore place the numerical boundary where the effective description is best controlled and evolve the kernel rather than repeatedly re-quantizing the system.

III. CMB OBSERVABLES FROM THE KERNEL

For the curvature perturbation,

$$\langle \zeta_{\mathbf{k}} \zeta_{\mathbf{k}'} \rangle = (2\pi)^3 \delta^3(\mathbf{k} + \mathbf{k}') P_\zeta(k), \quad (16)$$

$$\Delta_\zeta^2(k) = \frac{k^3}{2\pi^2} P_\zeta(k). \quad (17)$$

If all nonadiabatic modes have been integrated out,

$$P_\zeta(k) = \frac{\hbar}{2 \operatorname{Im} K_{\zeta, \text{eff}}(k)}. \quad (18)$$

The amplitude, tilt, and running follow from

$$A_s = \Delta_\zeta^2(k_*), \quad (19)$$

$$n_s - 1 = \left. \frac{d \ln \Delta_\zeta^2}{d \ln k} \right|_{k_*}, \quad (20)$$

$$\alpha_s = \left. \frac{d n_s}{d \ln k} \right|_{k_*}. \quad (21)$$

For the two graviton polarizations,

$$\Delta_T^2(k) = \frac{k^3}{2\pi^2} \sum_{\lambda=+, \times} P_{h_\lambda}(k), \quad (22)$$

where each covariance is obtained from the inverse tensor block of K_I . The tensor-to-scalar ratio and tensor tilt are

$$r(k) = \frac{\Delta_T^2(k)}{\Delta_\zeta^2(k)}, \quad n_T = \frac{d \ln \Delta_T^2}{d \ln k}. \quad (23)$$

These quantities are especially useful compactification diagnostics because internal geometry can modify the scalar and tensor blocks differently. A shift in A_s can often be absorbed into a change of the inflationary potential. A scale-dependent r , extra tensor polarizations, or a violation of $r = -8n_T$ is harder to imitate.

For rapid screening we use

$$A_s \simeq 2.1 \times 10^{-9}, \quad 0.955 < n_s < 0.980, \quad r_{0.05} < 0.036, \quad (24)$$

consistent with Planck and BICEP/Keck analyses [7–9]. These are not a substitute for a likelihood analysis; they are pass/fail benchmarks.

A. Recovery of the standard result

The reduced quadratic action for a single adiabatic mode is

$$S_\zeta^{(2)} = \frac{1}{2} \int d\eta \frac{d^3 k}{(2\pi)^3} z^2 (|\zeta'_k|^2 - c_s^2 k^2 |\zeta_k|^2), \quad (25)$$

where $z^2 = 2a^2 \epsilon \bar{M}_P^2 / c_s^2$. For canonical slow roll, $c_s = 1$. In approximate de Sitter space, $a = -1/(H\eta)$, and the regular solution may be normalized as

$$u_k(\eta) = (1 - ik\eta) e^{ik\eta}. \quad (26)$$

The boundary kernel is $K_\zeta = z^2 u'_k / u_k$, giving

$$\operatorname{Im} K_\zeta(k, \eta) = \frac{2\epsilon \bar{M}_P^2}{H^2} \frac{k^3}{1 + k^2 \eta^2}. \quad (27)$$

At late times $|k\eta| \ll 1$,

$$\operatorname{Im} K_\zeta \longrightarrow \frac{2\epsilon \bar{M}_P^2}{H^2} k^3. \quad (28)$$

Equations (10) and (18) then give

$$\Delta_\zeta^2 = \frac{\hbar H_*^2}{8\pi^2 \epsilon_* \bar{M}_P^2}, \quad (29)$$

while the tensor block gives

$$\Delta_T^2 = \frac{2\hbar H_*^2}{\pi^2 \bar{M}_P^2}, \quad r = 16\epsilon_*. \quad (30)$$

The method therefore reproduces the textbook answer. Its value is not a different prediction in the decoupling limit, but a cleaner route for deciding when a compactification does not lie in that limit.

IV. HOW COMPACTIFICATION DATA ENTER

Consider a $D = 4 + d$ dimensional metric

$$ds_D^2 = e^{2A(\phi)} g_{\mu\nu}(x) dx^\mu dx^\nu + G_{mn}(y; \phi) dy^m dy^n, \quad (31)$$

where $\phi^a(x)$ are breathing, shape, and flux-supported moduli. In the four-dimensional Einstein frame,

$$S_4 = \int d^4x \sqrt{-g} \left[\frac{\bar{M}_P^2}{2} R - \frac{1}{2} G_{ab}(\phi) \partial_\mu \phi^a \partial^\mu \phi^b - V_{\text{eff}}(\phi) \right] + S_{\text{light}}, \quad (32)$$

Fluctuations on the internal manifold are expanded in eigenfunctions of the appropriate scalar, vector, spinor, and Lichnerowicz operators. For a scalar harmonic,

$$-\nabla_{X_d}^2 Y_I = \lambda_I Y_I, \quad \chi(x, y) = \sum_I \chi_I(x) Y_I(y). \quad (33)$$

The four-dimensional masses contain λ_I plus curvature, flux, and interaction contributions. The internal geometry therefore enters through the spectral data (λ_I, d_I) and overlap integrals that determine mixing.

Separate observable variables $L = (\zeta, h_+, h_-)$ from internal variables $H = (s^\alpha, \chi^r)$. The quadratic boundary kernel is

$$K = \begin{pmatrix} K_{LL} & K_{LH} \\ K_{HL} & K_{HH} \end{pmatrix}. \quad (34)$$

Integrating out unobserved internal variables gives

$$K_{L,\text{eff}} = K_{LL} - K_{LH} K_{HH}^{-1} K_{HL}. \quad (35)$$

This Schur complement is the compactification correction. It contains changes in normalization, scale-dependent mixing, resonant structure, and the onset of instability when an eigenvalue of K_{HH} approaches zero.

When an internal mode has mass $M \gg H$ and varies adiabatically,

$$\delta K_L \sim -\frac{K_{LH} K_{HL}}{M^2} = \mathcal{O}\left(\frac{H^2}{M^2}\right) K_{LL}, \quad (36)$$

up to couplings and slow-roll factors. Distinct manifolds then become observationally degenerate. If $M \sim H$, corrections can be large and may appear in power-spectrum features or higher-point functions [10].

For numerical work one may evolve the kernel directly. For a canonically normalized vector v obeying

$$v'' + \Omega^2(k, \eta) v = 0, \quad (37)$$

let F be a fundamental matrix of contour-selected solutions. The kernel is

$$K = F' F^{-1}, \quad (38)$$

up to the kinetic prefactor, and satisfies the Riccati equation

$$K' + K^2 + \Omega^2 = 0. \quad (39)$$

This avoids separately normalizing every mixed mode and is convenient for scanning flux sectors or manifolds.

V. WORKED EXAMPLE: $M_4 \times S^2 \times \mathbb{CP}^2$

A. Background and flux branches

Take the two-radius ansatz used in the UECKK and spin-torsion companion papers [4, 6],

$$ds_{10}^2 = e^{2A} g_{\mu\nu} dx^\mu dx^\nu + a_1^2 d\Omega_2^2 + a_2^2 ds_{\mathbb{CP}^2}^2, \quad (40)$$

where the reference Fubini-Study metric is normalized once and all powers of a_2 are explicit. The internal volume scales as

$$V_6 \propto a_1^2 a_2^4. \quad (41)$$

The two logarithmic radii may be written

$$u = \ln(a_1/a_{1,0}), \quad v = \ln(a_2/a_{2,0}). \quad (42)$$

These fields have the Einstein-frame kinetic metric derived in Appendix B. The important point is that unimodularity fixes the full spacetime measure, not the internal volume by itself. In the reduced phase-space treatment of unimodular quantum cosmology, for example, the unimodular constraint solves for the lapse in terms of the spatial volume while the spatial scale factor remains dynamical [11]. The ten-dimensional analogue is that a change of

$$\sigma = 2u + 4v = \ln(V_6/V_{6,0}) \quad (43)$$

can be compensated by the external determinant or lapse. Therefore the primary UECKK screening calculation retains the full two-radion metric. A fixed-internal-volume projection, with shape coordinate $\chi = u - v$, is still useful as an optional truncation, but it is not the generic consequence of unimodularity.

A useful flux-and-curvature truncation is

$$U(a_1, a_2) = -A \left[\frac{2}{a_1^4 a_2^4} + \frac{24}{a_1^2 a_2^6} \right] + B \left[\frac{m^2}{2a_1^6 a_2^4} + \frac{qn^2}{a_1^2 a_2^8} \right] - C \frac{N_{\text{fam}}^2}{a_1^2 a_2^4}. \quad (44)$$

The first line is the internal-curvature contribution in the chosen reduction convention, the second contains quantized flux, and the final term is a volume-dependent attractive contribution. The coefficients A, B, C, q are model parameters until derived from the microscopic action. The family index is

$$N_{\text{fam}} = m \frac{n^2 - 1}{8}. \quad (45)$$

Both (3, 3) and (1, 5) give $N_{\text{fam}} = 3$, but their flux terms are

$$U_{\text{flux}}^{(3,3)} = B \left[\frac{9}{2a_1^6 a_2^4} + \frac{9q}{a_1^2 a_2^8} \right], \quad (46)$$

$$U_{\text{flux}}^{(1,5)} = B \left[\frac{1}{2a_1^6 a_2^4} + \frac{25q}{a_1^2 a_2^8} \right]. \quad (47)$$

They therefore have different minima, modulus Hessians, and overlap tensors even though their net chiral family number is identical.

The first background test is algebraic:

$$\partial_{a_i} U = 0, \quad \text{eigenvalues } [G^{-1} \nabla \nabla U] > 0. \quad (48)$$

The first condition locates a compactification saddle; the second tests local stability in the homogeneous truncation. A negative eigenvalue rejects the branch before any CMB calculation.

B. Internal spectra

For a sphere of radius a_1 , scalar harmonics obey

$$\lambda_\ell^{S^2} = \frac{\ell(\ell+1)}{a_1^2}, \quad d_\ell^{S^2} = 2\ell+1. \quad (49)$$

For the standard Fubini–Study normalization on $\mathbb{CP}^2$,

$$\lambda_p^{\mathbb{CP}^2} = \frac{4p(p+2)}{a_2^2}, \quad d_p^{\mathbb{CP}^2} = (p+1)^3, \quad (50)$$

with a rescaling if a different curvature convention is used. Product harmonics have

$$\lambda_{\ell p} = \lambda_\ell^{S^2} + \lambda_p^{\mathbb{CP}^2}. \quad (51)$$

Flux shifts the spectra of charged fields and changes the internal wave functions. That shift is essential for distinguishing (3,3) from (1,5): the branches have the same index but not the same mode profiles or mixing integrals.

C. Quadratic kernel

Retain the adiabatic mode, the two homogeneous moduli, and a finite set of KK modes,

$$q = (\zeta, u, v, h_+, h_-, \chi_{\ell p}^{(r)}). \quad (52)$$

After solving the four-dimensional lapse and shift constraints, the scalar block has the schematic action

$$S_S^{(2)} = \frac{1}{2} \int d\eta \frac{d^3 k}{(2\pi)^3} [Q'^T Q' - Q^T \Omega_S^2(k, \eta) Q], \quad (53)$$

where Q is the canonically normalized scalar vector. The matrix Ω_S^2 contains the field-space curvature, the canonically normalized Hessian of U , the KK eigenvalues, and mixing terms proportional to the turning rate of the background trajectory. The contour-selected fundamental matrix F_S gives

$$K_S = F_S' F_S^{-1}. \quad (54)$$

The effective curvature kernel follows from Eq. (35). The tensor block is computed independently, including any

mixing of the four-dimensional graviton with internal transverse-traceless metric modes.

For a rapid branch comparison, define

$$\mathcal{D}_{(m,n)}(k) = \{\text{eig } K_I, A_s, n_s, \alpha_s, r, n_T, \beta_{\text{iso}}\}_{(m,n)}. \quad (55)$$

The two branches are distinguishable only if

$$\mathcal{D}_{(3,3)}(k) \neq \mathcal{D}_{(1,5)}(k) \quad (56)$$

by more than observational and theoretical uncertainties. If both modulus masses satisfy $M_u, M_v \gg H$ and all charged KK modes are heavier still, both branches reduce to the universal single-field kernel. The method then returns a useful null result: the CMB cannot discriminate the flux distribution at quadratic order.

D. What can make the branches differ

There are four leading possibilities. First, one branch may be unstable. Second, one may have a modulus with $M \sim H$, producing isocurvature or a scale-dependent Schur correction. Third, the time dependence of $V_6 \propto a_1^2 a_2^4$ may modify the tensor normalization through an evolving effective Planck mass. Fourth, flux-dependent KK levels may lie near H and leave oscillatory or non-Gaussian signatures. The scalar power-law spectrum alone is unlikely to identify the branch if both are deep in the decoupling regime.

VI. $M_4 \times S^3 \times S^3$ AS A CONTROL

Consider

$$ds_{10}^2 = e^{2A} g_{\mu\nu} dx^\mu dx^\nu + b_1^2 d\Omega_{3,1}^2 + b_2^2 d\Omega_{3,2}^2. \quad (57)$$

This product has the same number of internal dimensions as $S^2 \times CP^2$ but different curvature and spectral data. The internal volume scales as $b_1^3 b_2^3$. Two logarithmic radii again give a volume and a shape mode.

For a unit three-sphere the scalar harmonics satisfy

$$\lambda_\ell^{S^3} = \frac{\ell(\ell+2)}{b^2}, \quad d_\ell^{S^3} = (\ell+1)^2. \quad (58)$$

Product eigenvalues are

$$\lambda_{\ell_1 \ell_2} = \frac{\ell_1(\ell_1+2)}{b_1^2} + \frac{\ell_2(\ell_2+2)}{b_2^2}. \quad (59)$$

A stabilizing model may include curvature, a cosmological term, and quantized three-form flux on each factor. Its Einstein-frame potential can be written generally as

$$U_{33}(b_1, b_2) = U_{\text{curv}} + U_{\text{flux}} + U_{\text{other}}, \quad (60)$$

with each term carrying a fixed power of b_1 and b_2 determined by dimensional reduction. The kernel algorithm is

unchanged: locate a stable saddle, canonically normalize the two moduli, construct Ω^2 , solve the regular matrix mode problem, and take the Schur complement.

This control example makes the main methodological point. The observable formulas do not need to be reinvented. What changes is the internal input:

$$\{G_{ab}, U, \lambda_I, d_I, \text{overlaps}\}_{S^2 \times \mathbb{CP}^2} \longrightarrow \{G_{ab}, U, \lambda_I, d_I, \text{overlaps}\}_{S^3 \times S^3}. \quad (61)$$

The same code then produces $K_{\zeta, \text{eff}}$ and $K_{T, \text{eff}}$. A proposed universe is rejected if either kernel fails the criteria below. If both manifolds pass and their internal modes decouple, their CMB two-point spectra are not a useful discriminant; one must turn to particle content, non-Gaussianity, or other observables.

VII. NUMERICAL DEMONSTRATION OF THE SCREENING PIPELINE

The preceding sections define a method, but a useful screening paper should also show that the chain from compactification data to an observational curve is operational. We therefore give a transparent benchmark calculation. The chosen dimensionless coefficients produce stable saddles and expose each computational step. They are not yet microscopic predictions of either compactification; a genuine verdict on a manifold requires coefficients and mixing matrices derived from its higher-dimensional action.

For a product of two Einstein spaces of dimensions d_1 and d_2 , with logarithmic radii $\beta_i = \ln a_i$, the unconstrained Einstein-frame kinetic matrix in the homogeneous truncation is

$$G_{ij} = \bar{M}_P^2 \left(d_i \delta_{ij} + \frac{1}{2} d_i d_j \right). \quad (62)$$

Thus, in units $\bar{M}_P = 1$,

$$G_{S^2 \times \mathbb{CP}^2} = \begin{pmatrix} 4 & 4 \\ 4 & 12 \end{pmatrix}, \quad G_{S^3 \times S^3} = \begin{pmatrix} 15/2 & 9/2 \\ 9/2 & 15/2 \end{pmatrix}. \quad (63)$$

We keep the full two-radii matrix. In the UECKK interpretation unimodularity fixes the total spacetime measure, not V_6 alone. Appendix B records the metric, and Appendix C gives the coordinate-invariant $G^{-1}H$ construction used in the main-text decoupling table. For $S^2 \times \mathbb{CP}^2$ we set $A = B = q = 1$ and $C = 0.05$ in Eq. (44). For the control model we use

$$U_{33} = -6 (b_1^{-5} b_2^{-3} + b_1^{-3} b_2^{-5}) + \frac{9}{2} (b_1^{-9} b_2^{-3} + b_1^{-3} b_2^{-9}) + 0.2 b_1^{-3} b_2^{-3}. \quad (64)$$

The extrema are found in logarithmic variables and the canonical mass matrix is $G^{-1} \nabla \nabla U$. Table I lists the results. All three examples possess a stable homogeneous

TABLE I. Dimensionless benchmark minima and eigenvalues of $G^{-1} \nabla \nabla U$. These values demonstrate the pipeline and are not microscopic predictions.

Model	a_1 or b_1	a_2 or b_2	m_-^2	m_+^2
(3, 3) on $S^2 \times \mathbb{CP}^2$	0.929	0.725	122.6	255.6
(1, 5) on $S^2 \times \mathbb{CP}^2$	0.535	1.151	34.1	72.1
$S^3 \times S^3$ control	1.033	1.033	12.1	30.3

TABLE II. Canonical homogeneous-modulus masses for the benchmark $S^2 \times \mathbb{CP}^2$ branches. The optional column $m_{\text{fixed } V_6}^2$ is the one-field mass obtained only after imposing the additional fixed-internal-volume truncation. All entries use the same dimensionless potential units as Table I.

branch	a_1	a_2	m_-^2	m_+^2	$m_{\text{fixed } V_6}^2$
(3, 3)	0.929	0.725	122.6	255.6	241.6
(1, 5)	0.535	1.151	34.1	72.1	72.0

saddle in this benchmark, but their radii and canonical mass eigenvalues differ substantially.

The first CMB-scale use of these numbers is the decoupling test. At a stationary point the covariant mass matrix is

$$(M^2)^I{}_J = G^{IK} \nabla_K \nabla_J U = G^{IK} H_{KJ}, \quad H_{KJ} = \partial_K \partial_J U, \quad (65)$$

because the gradient of U vanishes and the logarithmic-radiation metric is constant. The dimensionless quantities that enter the per- k kernel screen are

$$\mu_{\pm}^2 = \frac{m_{\pm}^2}{H_*^2}. \quad (66)$$

Table II is therefore a main-text result: it gives the canonical masses consumed by the minimal per- k implementation. The fixed- V_6 column is retained only as a diagnostic one-field truncation; it is not the default UECKK interpretation of unimodularity.

A. Observed amplitude and the tensor constraint

Combining $A_s = H_*^2 / (8\pi^2 \epsilon_* \bar{M}_P^2)$ with $r = 16\epsilon_*$ and $\bar{M}_P = \sqrt{8\pi} M_P$ gives

$$\frac{H_*}{M_P} = \frac{1}{4} \sqrt{\pi A_s r}. \quad (67)$$

Figure 2 uses $A_s = 2.10 \times 10^{-9}$ and the BICEP/Keck limit $r_{0.05} < 0.036$ [8, 9]. In the canonical single-field limit the bound corresponds to $H_*/M_P < 3.9 \times 10^{-6}$. A proposed compactification supplies the background value of H_* and the tensor kernel supplies r ; their point must lie on this curve.

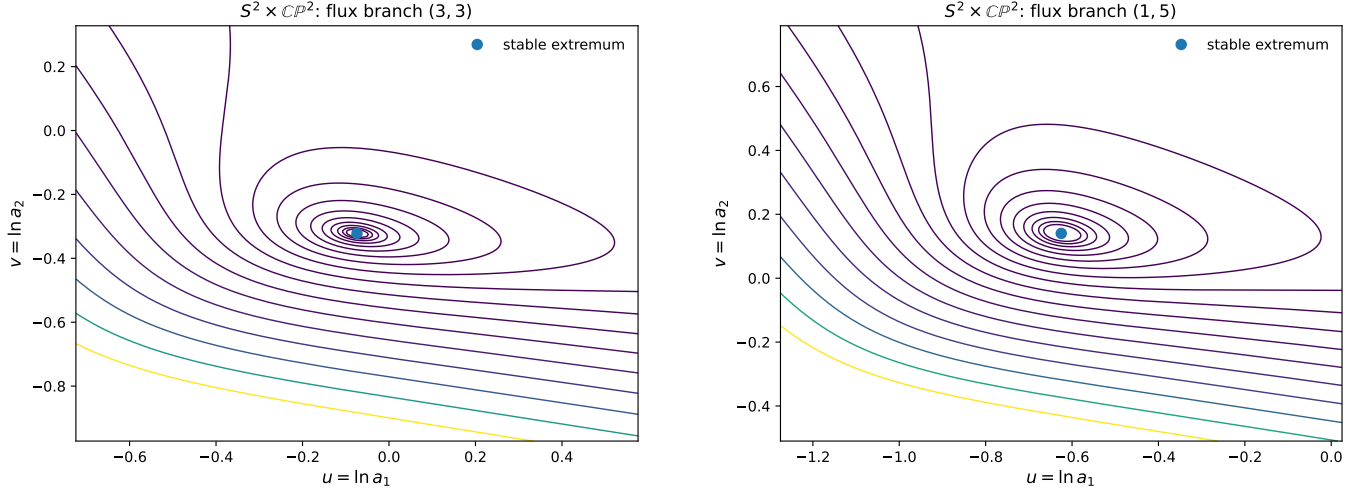


FIG. 1. Benchmark potentials for the two $S^2 \times \mathbb{CP}^2$ branches. Left: (3,3); right: (1,5). The dots mark the stable extrema. Although the two branches have the same family index, their flux distributions move the minimum and change the local Hessian.

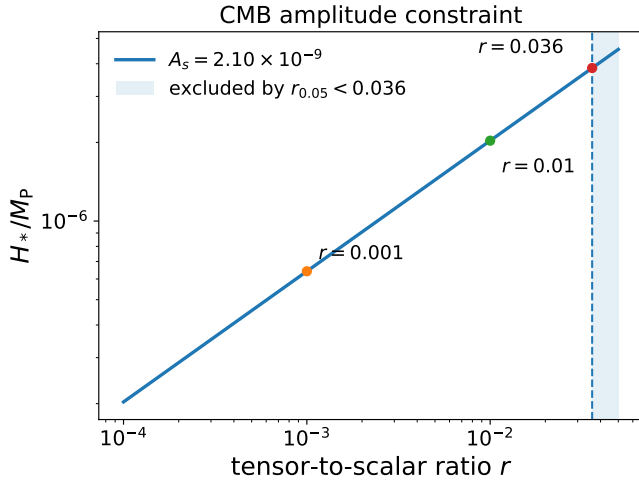


FIG. 2. The Hubble scale required by the measured scalar amplitude as a function of r . The shaded region is excluded by the current BICEP/Keck bound.

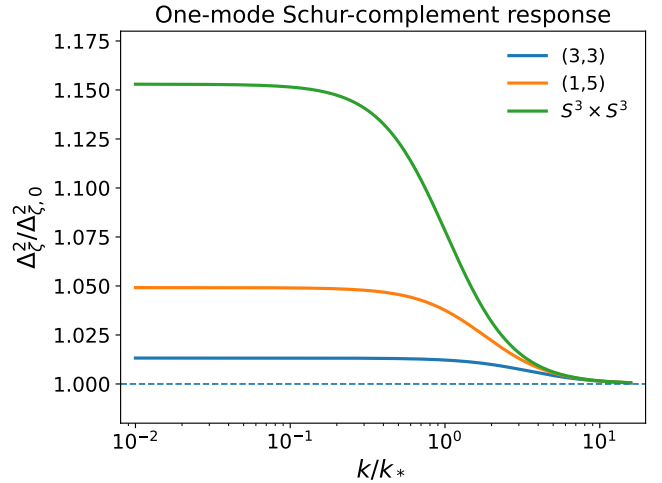


FIG. 3. Illustrative scalar-power response from one internal mode. The different mass eigenvalues generate visibly different scale dependences.

B. A one-mode Schur-complement visualization

To display the last algebraic step, retain one internal scalar with dimensionless mass $\mu = M/H$ and approximate the imaginary kernel near the pivot by

$$\frac{\text{Im } K_{\zeta, \text{eff}}}{\text{Im } K_{\zeta, 0}} = 1 - \frac{\gamma^2}{\mu^2 + (k/k_*)^2}. \quad (68)$$

Since the power is inversely proportional to $\text{Im } K$, the response is

$$\frac{\Delta_{\zeta}^2}{\Delta_{\zeta, 0}^2} = \left[1 - \frac{\gamma^2}{\mu^2 + (k/k_*)^2} \right]^{-1}. \quad (69)$$

For Fig. 3 we set the common illustrative values $H^2 = 10$ and $\gamma = 0.4$, and insert the lighter canonical mass from Table I. The calculation is deliberately simple enough to reproduce by inspection. A full model replaces Eq. (68) by the Riccati evolution of the constrained matrix kernel.

C. Direct comparison with the measured scalar spectrum

Figure 4 carries the benchmark through to an observational screen. The band is the pointwise envelope generated by the Planck 2018 constraints $n_s = 0.9649 \pm 0.0042$ and $\ln(10^{10} A_s) = 3.044 \pm 0.014$ [7, 12]. Each benchmark curve is normalized to the measured amplitude at

$k_* = 0.05 \text{ Mpc}^{-1}$; the test therefore probes the additional scale dependence produced by the internal mode rather than refitting the amplitude.

With the illustrative choices above, the (3,3) curve remains inside this one-standard-deviation parameter envelope over about 94% of the plotted logarithmic k grid, the (1,5) curve over about 28%, and the $S^3 \times S^3$ control over about 5%. These percentages are not likelihoods and must not be interpreted as observational confidence levels. They demonstrate that the kernel method can return a positive or negative screening verdict once the compactification masses and mixings are supplied. In this benchmark the (3,3) branch is close to the observed power-law shape, whereas the lighter internal modes of the other examples generate excessive low- k curvature.

The comparison can be implemented literally as a loop over wavenumber. For the minimal algebraic screen one sets $x = k/k_*$ and uses

$$R_{\mathcal{M}}(k) = \left[1 - \mathbf{b}^T (\text{diag } \mu_a^2 + x^2 \mathbf{1})^{-1} \mathbf{b} \right]^{-1}, \quad \mu_a^2 = \frac{m_a^2}{H_*^2}, \quad (70)$$

where m_a^2 are the canonical hidden-sector eigenvalues and $\mathbf{b}$ is the dimensionless vector of curvature–modulus mixing strengths. The primordial curve is then normalized at the pivot,

$$\Delta_{\zeta, \mathcal{M}}^2(k) = \Delta_{\zeta, \text{Pl}}^2(k) \frac{R_{\mathcal{M}}(k)}{R_{\mathcal{M}}(k_*)}. \quad (71)$$

Equations (70)–(71) are the shortest useful implementation of the LBK screen for each k . They do not replace the full Riccati evolution; rather, they provide a reproducible first pass that consumes the compactification masses and an assumed mixing vector before a full background trajectory is known. The source package includes the script `minimal_per_k_kernel.py`, which evaluates these equations on a logarithmic k grid and writes Fig. 5.

D. Matrix-evolved implementation of the per- k workflow

The algebraic screen above omits propagation. We therefore also implement the matrix mode evolution listed in Appendix D. The implementation retains one observable canonical mode and two hidden canonical modes. For each k it evolves a fundamental matrix F in the variable $x = -k\eta$,

$$F_{,xx} + \left[\mathbf{1} + \frac{C(x, k)}{x^2} \right] F = 0, \quad (72)$$

from the regular positive-frequency condition $F \propto e^{ix} \mathbf{1} / \sqrt{2k}$. The illustrative matrix C contains the two canonical mass ratios and a localized curvature–modulus turn. The turn is fixed in conformal time, so different wavenumbers encounter it at different values of x and acquire a scale-dependent response.

TABLE III. Diagnostics from the matrix-evolved demonstration. The effective tilt is fitted over $0.01 \leq k \leq 0.1 \text{ Mpc}^{-1}$. The final column is the largest absolute fractional departure from the pivot-normalized Planck power law over the plotted interval.

Model	inside [%]	n_s^{eff}	max. residual
(3,3)	67.8	0.9695	0.133
(1,5)	47.8	0.9811	0.647
$S^3 \times S^3$	61.1	0.9733	0.344

At the final boundary the code constructs

$$K(k) = k F_{,x} F^{-1}, \quad K_{\zeta, \text{eff}} = K_{\zeta\zeta} - K_{\zeta H} K_{HH}^{-1} K_{H\zeta}, \quad (73)$$

checks every eigenvalue of the Hermitian imaginary part of K , and compares $\text{Im } K_{\zeta, \text{eff}}$ with an otherwise identical zero-mixing evolution. The resulting spectrum is normalized at the pivot before comparison with the Planck power law.

For all three demonstrations the minimum eigenvalue of the imaginary kernel remains positive over the complete grid. This verifies that the chosen contour and toy evolution define a normalizable Gaussian state. The resulting percentages and tilts are not observational statements about the compactifications: they depend on the deliberately chosen turn profile. Their purpose is narrower but concrete—every step of the proposed Appendix-D loop has now been executed in code.

The final qualification is essential. The present plot proves that the procedure works and shows which derived quantities control the answer. It does not yet provide positive evidence for the manifold, because changing the benchmark mixing changes the curves. A fourth companion paper on $M_4 \times S^2 \times \mathbb{CP}^2$ would become possible when G_{ab} , the full quadratic mixing matrix, and the inflationary trajectory are derived from the same microscopic model used in the family, torsion, and stability papers.

VIII. A PRACTICAL CMB ACCEPTABILITY TEST

The method is a staged filter. First derive the four-dimensional Einstein-frame background, require positive kinetic terms, controlled stabilization, and a KK gap consistent with the truncation. Next evolve the constrained scalar and tensor kernels over the observable range and require

$$\text{eig } K_I(k) > 0 \quad (74)$$

for every retained physical mode. After integrating out hidden modes, compute $A_s, n_s, \alpha_s, r, n_T$, and the isocurvature fraction, and apply broad observational cuts before running a full Boltzmann likelihood. Two compactifications are distinguishable only if their predicted spectra

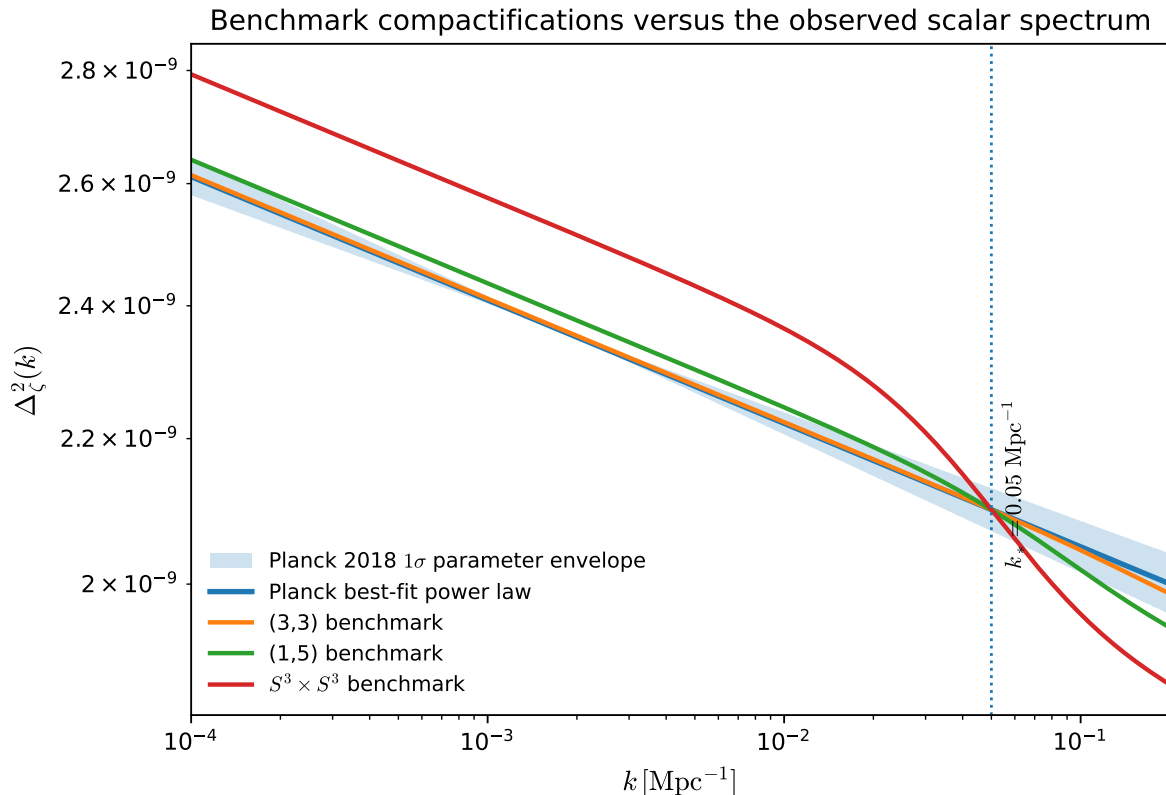


FIG. 4. Benchmark primordial spectra compared with the Planck 2018 best-fit power law and its one-standard-deviation parameter envelope. This is an observational comparison, but not yet a manifold prediction: the benchmark mixing γ and the overall potential normalization have not been derived from either higher-dimensional action. A genuine confirmation or exclusion of $M_4 \times S^2 \times \mathbb{CP}^2$ requires those microscopic inputs.

differ by more than the combined theoretical and observational uncertainty; otherwise the correct conclusion is decoupling or CMB degeneracy.

IX. DISCUSSION

The Gaussian wave-functional width and the single-field spectrum are standard; the proposed contribution is their use as a common screening object for higher-dimensional models. Positivity tests stability before transfer functions are computed, the Schur complement exposes decoupling, and matrix evolution handles coupled modes uniformly. The method is useful only insofar as this organization rejects models or identifies the next observable more efficiently than a full conventional pipeline.

The CMB will not generally determine a unique manifold. Distinct compactifications can share the same light theory, and viable scalar spectra must already be close to a featureless power law. Realistic outcomes are exclusion of a branch, constraints on modulus masses and couplings, or distinction through tensors, isocurvature, non-Gaussianity, or massive exchange. The (3, 3) and (1, 5) branches illustrate this point: their family indices

agree, while flux support, mode profiles, and kernels differ; if all internal masses satisfy $M \gg H$, the difference disappears at quadratic order.

The construction is semiclassical and assumes a controlled saddle, contour, and truncation. Moving an intermediate boundary only repartitions state preparation and propagation, and a high-curvature starting surface is not required. Passing the two-point screen does not guarantee successful reheating, late-time stability, acceptable non-Gaussianity, or particle physics, especially near the compactification scale where higher-derivative terms and the full KK tower may matter.

X. CONCLUSION

A higher-dimensional cosmology can be screened through the imaginary part of its constrained Lorentzian boundary kernel, which gives the covariance of the physical perturbations and encodes mixing with internal modes. The method reproduces standard single-field inflation in the decoupling limit and applies unchanged when the compactification is changed; only the field-space, mass, mixing, and spectral inputs differ.

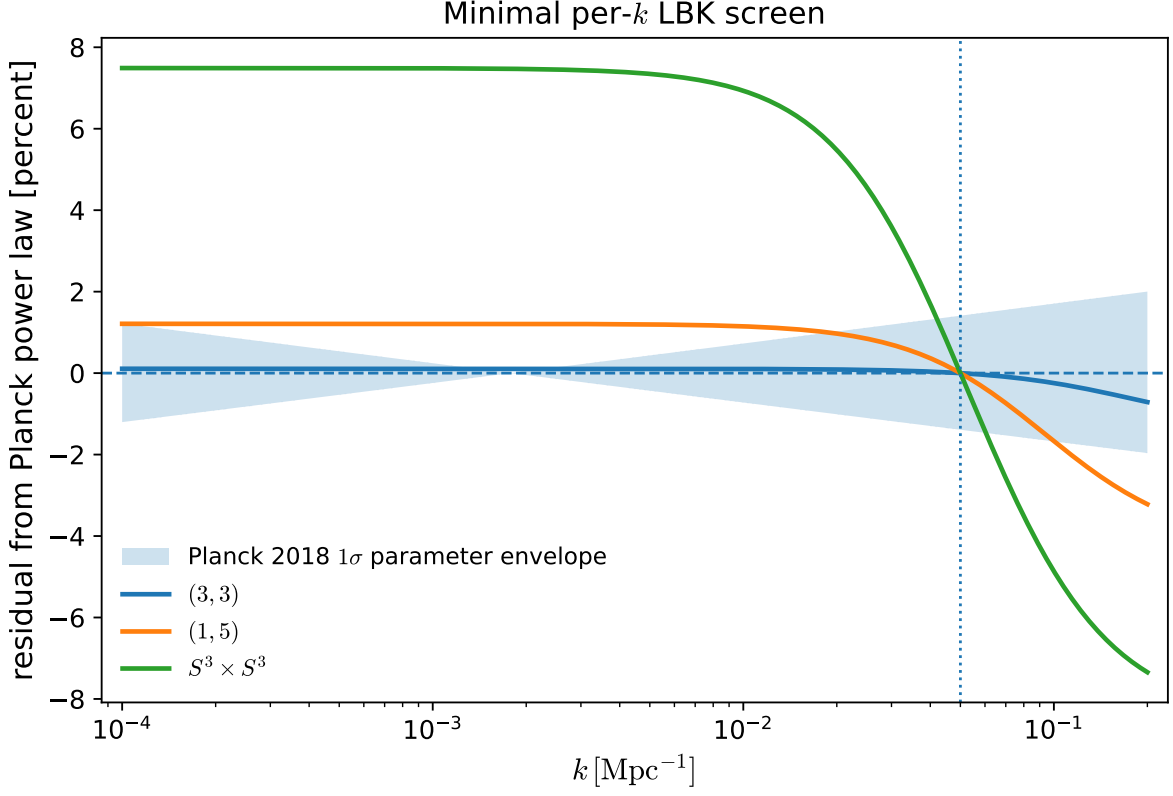


FIG. 5. Minimal per- k LBK implementation. The curves show the fractional residual relative to the Planck best-fit primordial power law after pivot normalization. The shaded region is the pointwise envelope generated by the 1σ uncertainties in A_s and n_s . For the illustrative choices $H^2 = 10$ and $\mathbf{b} = (0.40, 0.22)$, the fractions of the plotted k grid inside the envelope are 93.2% for $(3, 3)$, 22.3% for $(1, 5)$, and 4.7% for $S^3 \times S^3$. This is a code-level demonstration of the screening workflow, not a final Planck likelihood.

TABLE IV. Minimal inputs and outputs of the boundary-kernel screening test.

Stage	Required input	Immediate output
Background	G_{ab} , V_{eff} , flux integers	stable saddle, $H(t)$, modulus masses
Internal spectrum	Laplace/Lichnerowicz eigenvalues and overlaps	KK masses, degeneracies, mixing matrices
Quadratic dynamics	constrained scalar and tensor actions	matrix frequencies Ω_S^2 , Ω_T^2
Lorentzian state	regularity or $i\epsilon$ contour	positive imaginary boundary kernel
CMB screen	$K_{\zeta, \text{eff}}$, $K_{T, \text{eff}}$	$A_s, n_s, \alpha_s, r, n_T, \beta_{\text{iso}}$
Discrimination	two or more candidate kernels	exclusion, distinguishability, or decoupling bound

Its role is an acceptability filter. It can reject unstable branches, identify light internal modes, quantify decoupling, and decide whether a full CMB likelihood calculation is warranted. Unique reconstruction of a manifold from the scalar spectrum is unlikely; useful discrimination will more plausibly come from tensors, isocurvature, non-Gaussianity, or violations of four-dimensional consistency relations.

Appendix A: Gaussian integration and the Schur complement

Let a Gaussian state depend on observed variables x and hidden variables y ,

$$\Psi(x, y) \propto \exp \left[\frac{i}{2\hbar} \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} A & B \\ B^T & C \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \right]. \quad (\text{A1})$$

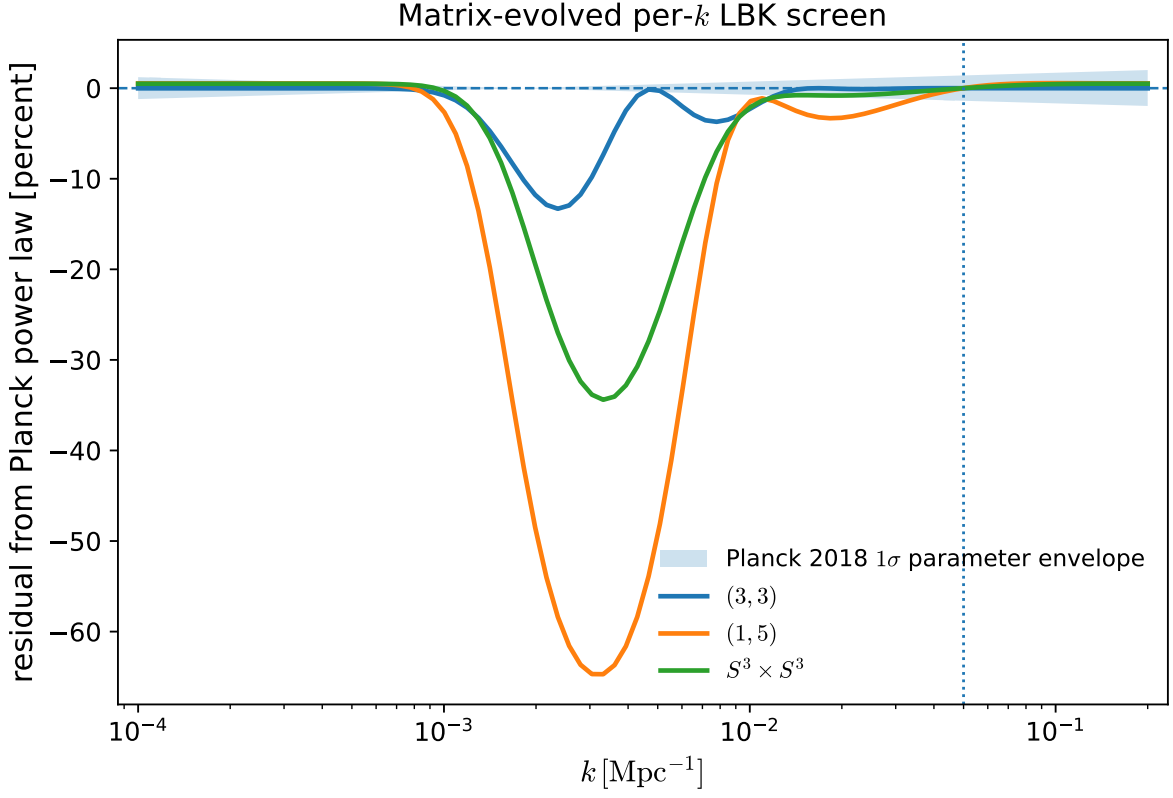


FIG. 6. Dynamical matrix-mode implementation of the per- k workflow. Unlike Fig. 5, these curves follow from numerical evolution of a coupled fundamental matrix, construction of the late-time boundary kernel, a positivity check, and a Schur complement. The localized turn and its mixing strengths remain illustrative rather than UECKK-derived.

Integrating over y gives

$$\Psi_{\text{eff}}(x) \propto (\det C)^{-1/2} \exp \left[\frac{i}{2\hbar} x^T (A - BC^{-1}B^T) x \right], \quad (\text{A2})$$

with the contour chosen so that the Gaussian converges. This proves Eq. (35). For a field theory the matrices carry momentum labels and functional determinants replace ordinary determinants.

Appendix B: Radion metric input for $S^2 \times \mathbb{CP}^2$ and the role of unimodularity

The full Einstein-frame reduction of the homogeneous $S^2 \times \mathbb{CP}^2$ radii is part of the UECKK compactification analysis and is given in the companion construction [4]. Here we record only the result and the short derivation needed to make the LBK screening calculation self-contained.

Let

$$u = \ln a_1, \quad v = \ln a_2, \quad (\text{B1})$$

where a_1 and a_2 are the radii of S^2 and $\mathbb{CP}^2$. For a product compactification with internal dimensions d_i , the four-dimensional Einstein-frame radion metric in logarithmic

coordinates is

$$\mathcal{G}_{ij} = d_i \delta_{ij} + \frac{d_i d_j}{D_4 - 2} = d_i \delta_{ij} + \frac{1}{2} d_i d_j, \quad D_4 = 4. \quad (\text{B2})$$

Thus, for $d_1 = 2$ and $d_2 = 4$,

$$\mathcal{G}_{uv} = \begin{pmatrix} 4 & 4 \\ 4 & 12 \end{pmatrix}, \quad G_{uv} = \bar{M}_P^2 \begin{pmatrix} 4 & 4 \\ 4 & 12 \end{pmatrix}. \quad (\text{B3})$$

The corresponding field-space line element is

$$ds_{\text{rad}}^2 = 4du^2 + 8dudv + 12dv^2. \quad (\text{B4})$$

In volume and shape variables,

$$\sigma = 2u + 4v, \quad \chi = u - v, \quad (\text{B5})$$

one obtains

$$ds_{\text{rad}}^2 = \frac{2}{3} d\sigma^2 + \frac{4}{3} d\chi^2. \quad (\text{B6})$$

Unimodularity fixes the full spacetime volume element, not the internal volume alone. In reduced unimodular cosmology it fixes Na^3 while the scale factor remains dynamical [11]; analogously, changes in $a_1^2 a_2^4$ may be compensated by the external determinant or lapse. The LBK screen therefore uses the full metric in Eq. (B3); fixed internal volume would be an additional truncation.

Appendix C: Canonical modulus masses and the first CMB-scale screen

The first compactification-level screen compares physical homogeneous-modulus masses with the inflationary scale. The coordinate-invariant construction uses the full metric in Eq. (B3).

Let

$$U(u, v) = \sum_A c_A \exp(-\alpha_A u - \beta_A v). \quad (\text{C1})$$

The Hessian in logarithmic coordinates is

$$H_{IJ} = \partial_I \partial_J U = \sum_A c_A \begin{pmatrix} \alpha_A^2 & \alpha_A \beta_A \\ \alpha_A \beta_A & \beta_A^2 \end{pmatrix} \exp(-\alpha_A u - \beta_A v), \quad (\text{C2})$$

with signs already squared by the second derivative. At a stationary point, connection terms in $\nabla_I \nabla_J U$ vanish; since the metric is constant in (u, v) , the physical mass matrix is simply

$$(M^2)^I_J = G^{IK} H_{KJ} = \frac{1}{M_{\text{Pl}}^2} (\mathcal{G}^{-1})^{IK} H_{KJ}. \quad (\text{C3})$$

Its two eigenvalues m_-^2 and m_+^2 are the masses that should be compared with H_*^2 in the homogeneous-sector CMB screen.

For the illustrative coefficients used in Sec. VII, $A = B = q = 1$ and $C = 0.05$, the resulting two-field masses are reported in Table II in the main text, where they are used as the input to the per- k screening calculation. The CMB-scale decoupling variables are those of Eq. (66). A branch with both $\mu_\pm^2 \gg 1$ has no homogeneous-modulus isocurvature problem at quadratic order; the moduli decouple and the scalar spectrum approaches the universal four-dimensional result. A branch with one $\mu^2 \sim 1$ requires the full coupled kernel evolution because curvature-modulus mixing can generate scale dependence or residual isocurvature. A negative eigenvalue rejects the branch at the homogeneous level. After the microscopic normalization of U is fixed by the UECKK action, Eq. (C3) gives the physical ratios $m_\pm^2/H_*^2$.

Appendix D: Implemented numerical workflow

The source package contains two executable levels of the screen. The script `minimal_per_k_kernel.py` evaluates the algebraic Schur-complement approximation in Eqs. (70)–(71). The script `matrix_per_k_kernel.py` follows through on the dynamical workflow and produces Fig. 6, Table III, a CSV table, and the L^AT_EX macros used in the main text.

For each wavenumber the code reads the canonical masses, assembles the coupled frequency matrix, initializes a positive-frequency fundamental matrix at $x_i = 60$, and evolves it to $x_f = 0.05$. It then forms $K = kF_{,x}F^{-1}$, checks positivity of $K_I = (K - K^\dagger)/(2i)$, integrates out hidden modes by the Schur complement, repeats with zero mixing, pivot-normalizes the response, and writes spectra and diagnostics over $10^{-4} \leq k \leq 0.2 \text{ Mpc}^{-1}$.

The illustrative frequency correction is

$$C(x, k) = \begin{pmatrix} -2 & g_1 T(x, k) & g_2 T(x, k) \\ g_1 T(x, k) & \mu_1^2 - 2 & 0 \\ g_2 T(x, k) & 0 & \mu_2^2 - 2 \end{pmatrix}, \quad (\text{D1})$$

where

$$T(x, k) = \exp \left[-\frac{1}{2\sigma_T^2} \ln^2 \left(\frac{x/k}{|\eta_T|} \right) \right]. \quad (\text{D2})$$

The numerical demonstration uses $|\eta_T| = k_*^{-1}$, $\sigma_T = 0.55$, and $(g_1, g_2) = (0.05, 0.025)$. These values are exposed at the top of the script and are not fitted. Replacing Eqs. (D1)–(D2) by the constrained scalar or tensor matrix derived from a compactification leaves the rest of the code path unchanged.

A physical UECKK run still requires the background trajectory, canonical curvature–radion mixing, and retained KK overlaps from the higher-dimensional action. Replacing the toy frequency constructor supplies these inputs; a precision stage would then pass the scalar, tensor, and isocurvature spectra to CLASS or CAMB and evaluate a likelihood rather than the illustrative pointwise envelope.

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